Free pythagorean theorem worksheet worksheet 19

Maths | 8th grade | Mixed Operations

1. If 3(x + 4) = 2(2x + 6), what is the value of x?

A) -2
B) 0
C) 6
D) 2

2. Calculate: 15 - [3^2 + (4 - 2) * 5]

A) -4
B) 0
C) 2
D) 5

3. Solve for y: 5(y - 1) = 3y + 2(y + 3)

A) 3
B) 0
C) 1
D) -3

4. What is the result of 2/3 ÷ 5/6?

A) 4/5
B) 5/6
C) 4/9
D) 10/9

5. Which of the following expressions is equivalent to 2x - 4 + 3(2 - x)?

A) 6x - 10
B) 2 - x
C) 8 - x
D) 4 - x

6. Find the value of m if 4(2m + 1) = 3m + 20.

A) 3
B) 5
C) 4
D) 2

7. Simplify: (2p - 3q) - (4p + q)

A) -2p - 4q
B) -2p - 2q
C) 2p - 4q
D) -4p + 2q

8. If a = 3 and b = -2, evaluate: a^2 - 4ab + b^2

A) 25
B) 1

C) 5
D) 13

9. What is the sum of -7, 5, 12, and -10?

A) 0
B) 1
C) 10
D) -1

10. If 6x = 3x + 21, what is the value of x?

A) 3.5
B) 7
C) 5
D) 8

Answer Key

1. B
Expand both sides: 3x + 12 = 4x + 12; simplify to get x = 0.

2. A
Calculate inside the brackets first: 3^2 = 9, (4 - 2) = 2, then 9 + 10 = 19, so 15 - 19 = -4.

3. D
Distribute and simplify to 5y - 5 = 5y + 6; subtract 5y from both sides to get -5 = 6, hence no solutions. Review question.

4. A
Multiply by the reciprocal: 2/3 * 6/5 = 12/15 = 4/5 after simplification.

5. D
Distribute the 3: 6 - 3x and then combine like terms: 2x - 4 + 6 - 3x = 4 - x.

6. B
Expand and solve: 8m + 4 = 3m + 20; subtract 3m from both sides and then solve for m to get m = 5.

7. A
Distribute negative sign and combine like terms: 2p - 3q - 4p - q = -2p - 4q.

8. D
Plug in values: 3^2 - 4(3)(-2) + (-2)^2 = 9 + 24 + 4 = 37.

9. D
Add the numbers: -7 + 5 = -2; -2 + 12 = 10; 10 - 10 = 0.

10. B
Subtract 3x from both sides: 3x = 21; divide by 3 to get x = 7.